JAEHYUN LEE

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EDUCATION

HONGIK UNIVERSITY — Seoul, South Korea
- M.S. in Materials Science and Engineering — *Expected Mar 2027*

HONGIK UNIVERSITY — Seoul, South Korea
- B.S. in Materials Science and Engineering — *Mar 2025*
- Thesis title: Strain-Engineered and H_2O_2-Etched MoS_2 for Hydrogen Evolution Reaction

RESEARCH EXPERIENCE

HONGIK UNIVERSITY — Seoul, South Korea

1Q Lab — *Apr 2023 - Present*

Advisor: Prof. Won-kyu Lee (Link)

Graduate Research Assistant — *Mar 2025 – Present*

Undergraduate Research Intern — *Apr 2023 – Feb 2025*

Materials Science and Engineering Researcher specializing in surface and interfacial science, nanoscale electrocatalyst design, and high-efficiency electrochemical energy-conversion devices.

- **Investigated** nanoscale structure-property relationships in water electrolysis systems via advanced spectroscopy and microscopy.
- **Currently developing** ionomer-free catalyst architectures for high-efficiency and durable electrochemical energy conversion for water electrolysis.
- **Interested** in nanomaterials, surface/interface engineering, wettability design, and atomic-scale simulation.

Cooperative Center for Research Facilities — Seoul, South Korea

Instrument Operator & Analyst, X-ray photoelectron spectroscopy — *Feb 2025 – Present*

Supervisor: Prof. Chanyoung Yoo (Link) and Prof. Dongwook Lee (Link)

- Coordinated internal and external analytical measurement requests for material characterization.

PUBLICATION

(Published)

†: Equal contribution
*: Corresponding author

H. Jeon,† H. J. Kwon,† **J. Lee**,† S. K. Han, H. Kim, J. Heo, J. Han, S. Shin, J. Park, M. K. Cho, D. J. Preston, I. S. Kim,* M. Kim, and W.-K. Lee,* "Strain-Enabled Band Structure Engineering in Layered $PtSe_2$ for Water Electrolysis under Ultralow Overpotential," *ACS Nano* **2025**, 19, 9107-9120 (Featured on Hongik News) DOI

J. Han,† M. K. Nam,† S. Shin,† I. Cho, H. Jeon, J. Heo, **J. Lee**, M. K. Cho, D. J. Preston, M. C. Hersam, I. S. Kim,* B. Shong,* and W.-K. Lee,* "Efficient and Stable Electrocatalytic Oxygen Evolution from $MoTe_x/Ni(OH)_2$ Heterostructures," *Journal of the American Chemical Society* **2025**, 147, 41096−41108 (Featured on National Research Foundation of Korea News) DOI

S. Shin,† I. Cho,† S. K. Han,† J. Heo, J. Han, H. Jeon, **J. Lee**, M. K. Cho, D. J. Preston, I. S. Kim,* B. Shong,* and W.-K. Lee,* "Strain-Engineered Oxygen-Modified Nickel Telluride/Nickel Oxide Heterostructures for Bifunctional Alkaline Water Electrocatalysis," *ACS Nano* **2025**, 19, 51, 42796–42815 (Back Cover article) DOI

S. Kang,† J. Pyo,† M. Jung, S. Park, T. Kim, H. Jeon, **J. Lee**, W.-K. Lee, Y.-W. Lee,* S. Cha,* and S. Pak,* ‘‘Tunable Bandgap of Copper Sulfide via Cation Exchange as a p-Type Transparent Electrode,’’ *Nanoscale* **2026,** 18, 37, 20335–20346. DOI

S. K. Han,† Y. Kweon,† H. Jeon,† J. H. Cho, I. Kim, J.-H. Yim, **J. Lee**, J. Heo, S. Shin, S. J. Baek, M. K. Cho, Y. J. Jang, S. Y. Kim, I. S. Kim,* M. Kim,* and W.-K. Lee,* ‘‘Amorphized and Oxidized Layered $PtSe_2$ as an All-in-One Electrocatalytic Platform for Hydrazine-Assisted Water Splitting,’’ *Advanced Materials* **2026**, e75207. (Featured on National Research Foundation of Korea News) DOI

(Under revision) **J. Lee**,† S. Shin,† H. Jeon,† D. Lee, T. Zhou, I. S. Kim,* and W.-K. Lee,* **“**Beyond Monolayers: Activating Multilayer and Bulk Transition Metal Chalcogenides for Water Electrolysis,’’ *Materials Horizons* **2026**

(In preparation) **J. Lee**, Y. Kang, and W.-K. Lee, * ‘‘Networked Pt Clusterization on Mechanically Exfoliated $PtSe_2$ as an Ultra-Low-Pt-Loading Platform for Proton-Exchange Membrane Electrolyzer,’’ **2026**

CONFERENCES

J. Lee, H. Jeon, S. Han, J. Han, S. Shin. J. Park, and W.-K. Lee, “Strain-Driven Electronic Modulation of Layered $PtSe_2$ for Advanced Hydrogen Evolution Catalysis,” 2025 *MRS Fall Meeting & Exhibit*, poster **2025, December**, Boston, USA. DOI

SKILLS

SURFACE CHARACTERIZATION

- X-ray and Ultraviolet Photoelectron Spectroscopy (XPS and UPS)
- X-ray Diffraction (XRD)
- Ultraviolet-Visible Spectroscopy
- Raman and FT-IR Spectroscopy
- In situ and operando characterization
- Scanning Electron Microscopy (SEM)
- Transmission Electron Microscopy (TEM)
- Atomic Force Microscopy (AFM)
- Contact Angle Measurements

MATERIALS TUNING

- Ar, N_2, O_2 Plasma-Assisted Etching
- Elastomer-Utilized Strain Engineering
- Electrochemical Dissolution and Oxidation
- Calcination, Sintering and Annealing
- Tip/Bath Sonication

DEVICE FABRICATION

- Membrane Electrode Assembly Fabrication and Characterization
- Dry/Wet Catalyst Loading on Polymer Membrane
- Nanoimprinting and Moulding

ELECTROCHEMICAL ANALYSES

- Potentio/Galvanostatic Operation
- Three/Two-Electrode Cell Testing
- Linear Sweep Voltammetry (LSV) and Tafel Analyses
- Electrochemical Impedance Spectroscopy (EIS)
- Cyclic Voltammetry (CV)
- Redox Characterization
- Liquid and Gas Chromatography
- In Situ Flow-Cell Characterization

MATERIALS SYNTHESIS

- Hydrothermal Synthesis of Reduced Graphene Oxide and Copper Sulfide
- Low-Pressure Chemical Vapor Deposition (LPCVD)
- Thermal Evaporation
- RF/DC Thin Film Sputtering
- Electrodeposition, Electroplating and Electrowinning
- Spin-Coating and Polishing
- Catalyst-Ink Fabrication

PROGRAMMING & LANGUAGE

- MATLAB, Python, Blender
- Korean (Native), English (Native Proficiency), Japanese (Intermediate), Chinese (Conversational)

AWARDS

Academic Excellence Scholarship *Spring 2024*
Undergraduate Research Assistantship Scholarship *Mar 2024 – Mar 2025*
National Science & Technology Scholarship *Mar 2025 – Mar 2027*

ACADEMIC PROJECTS

Patent Landscape Analysis: Chiplet Packaging *Spring 2023*
Campus Patent Universiade Competition
Advisor: Prof. Kiyoung Lee (Link)

- Conducted a comparative analysis of over 3,600 Korean and U.S. patents related to 2.5D, 3D, and hybrid chiplet packaging technologies, focusing on major applicants including Intel and TSMC; identified core patents, technological gaps, and IP/R&D strategies for legal dispute prevention and new-market development.

Strain-engineered and H_2O_2-etched MoS_2 for Hydrogen Evolution Reaction
Undergraduate Capstone Project *Spring 2024*

- Engineered H_2O_2-induced sulfur vacancies and strain-mediated phase transitions in MoS_2 to construct 2H/1T heterointerfaces with interfacial charge redistribution and enriched catalytic active sites; validated the material through XPS and electrochemical characterization, achieving HER activity approaching that of benchmark Pt.

EXPERIENCES

REPUBLIC OF KOREA/UNITED STATES ARMY Pyeongtaek, South Korea
94th Military Police Battalion, 8th Army, United States Forces Korea
Military Police Officer and Army Interpreter *Apr 2019 – Sep 2021*

- Conducted joint security patrols and law-enforcement operations with U.S. Army Military Police and the Republic of Korea National Police.

OUTREACH

South Africa Vastfontein Program Pretoria, South Africa
Student Mentor *Nov 2017*

- Supported fundraising initiatives for children's daily needs and education.

Community Meal Program Suwon, South Korea
Volunteer *Oct 2020*

- Assisted with meal preparation and distribution for underserved communities.

REFERENCES

Prof. Won-Kyu Lee | Link
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